

Supporting Information for

Indirect reciprocity with dual private assessment

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Supporting Information Text

In the main text, we show how dual assessments facilitate opinion synchronization under private assessment, thereby facilitating the evolution of cooperation. For the sake of exposition, we have assumed that donors are assessed according to a subset of the leading-eight norms: L3, L5, L6, and L8. In the following, we first complement these main-text results in three ways. In **Section 1**, we describe simulation results when donors are assessed according to the remaining leading-eight norms and the secondary-sixteen norms. In **Section 2**, we derive Eq. [3] in the main text, which we used to explain why Recipient Image Scoring (RIS) is particularly effective when assessing recipients. Next, in **Section 3**, we present details of the recovery analysis for L6-RIS. In particular, we derive the entries of the relevant transition matrix, and we show that in large populations, the recovery time increases linearly with population size (which distinguishes L6-RIS from L6-base). This analysis is complemented by a recovery analysis from a random configuration of opinions in **Section 4**, demonstrating that the recovery time remains fast even when we have multiple disagreements in the population.

Furthermore, we show that the results reported in the main text are robust to several model extensions. In **Section 5**, we show that Lx-RIS is stable not only against ALLC and ALLD mutants but also against all possible mutants that differ in their action rules. In **Section 6**, we study a model with imperfect observation, where agents observe the action with probability q. The results change smoothly as q decreases, indicating that the patterns reported in this paper are not artifacts of perfect observation. Finally, in **Section 7**, we use a large-scale computation to enumerate all possible combinations of R_1, R_2, and S, and to evaluate the evolutionary stability of cooperation. We find that the most successful norms under private assessment combine a leading-eight R_1 rule, an RIS-like R_2 rule, and a DISC-like action rule (i.e., donors cooperate with good recipients and defect against bad ones).

1. Results for the remaining leading-eight and the secondary-sixteen norms

We first discuss simulation results when donors are assessed according to one of the four remaining leading-eight norms: L1, L2, L4, or L7. As shown in the main text **Table 1**, the two norms L1 and L2 are somewhat special among the leading-eight norms; they apply a different action rule. All other norms suggest that the donor should cooperate if and only if the recipient is deemed good,

$$S(G,G)=C, \quad S(G,B)=D, \quad S(B,G)=C, \quad S(B,B)=D. \qquad [1]$$

In contrast, L1 and L2 suggest that the donor should always cooperate unless the donor is good and the recipient is bad,

$$S(G,G)=C, \quad S(G,B)=D, \quad S(B,G)=C, \quad S(B,B)=C. \qquad [2]$$

For better comparison with the other norms, we therefore report results for two variants of these norms: the original variants L1 and L2 with action rule Eq. [2], and modified variants L1v and L2v with action rule Eq. [1]. Overall, we consider six norms for assessing donors in this section: L1, L1v, L2, L2v, L4, and L7.

Fig. S1 is the analogue of the main text **Fig. 3**. Here, for each of the six donor norms, we consider all possible recipient-assessment rules R_2. For each resulting pair, we again report the self-cooperation level of the respective norm (x-axis) and the critical benefit-to-cost ratio above which the norm is stable against ALLD (y-axis). Norms in the lower-right corner of a given panel are both cooperative and particularly resistant to invasion by defectors. The overall trend is similar to the results reported in the main text: norms with R_2 = RIS are typically very strong both in terms of self-cooperation level and stability against ALLD.

Fig. S2 is the analogue of **Fig. 5**. We consider twelve different focal norms by combining each of the six donor-assessment rules with either R_2 = base or R_2 = RIS for assessing recipients. In each case, we study how well the respective norm performs in an evolutionary competition with ALLC and ALLD. The overall trend is similar to before: norms with R_2 = RIS generally outperform the base norms. Interestingly, however, for L1 and L2 we observe the opposite. Because both norms prescribe $S(B,B)=C$, donors with bad reputations tend to cooperate with recipients with bad reputations. In that case, observers who use the RIS rule would infer the recipient to be good. Hence, the RIS rule may facilitate the invasion of ALLD mutants once mutants are sufficiently common. In line with this view, we note that for both L1v and L2v we recover the previous pattern, with RIS being more successful than the base norm. These results suggest that the RIS rule for assessing recipients is most effective when it matches the donor's action rule in Eq. [1].

Fig. S3 systematically explores the evolutionary competition among the focal norm, ALLC, and ALLD, where the focal norm consists of one of the six donor-assessment rules and an arbitrary recipient-assessment rule. This figure is analogous to the main text **Fig. 6**, and it supports a similar conclusion. For L1v, L2v, L4, and L7, the RIS rule is again among the most effective of all possible rules for judging recipients.

For completeness, we also report corresponding results for the 'secondary-sixteen' norms (1). The secondary-sixteen are another class of norms that realize full cooperation and form strict Nash equilibria under public assessment for $b/c > 2$. Together with the leading-eight, they exhaust the deterministic norms that can sustain cooperation under public assessment. The most distinctive feature of the secondary-sixteen is that they are more context-sensitive than the leading-eight, in the sense that they prescribe different actions and reputational assessments for different interaction contexts. More specifically, they prescribe defection when a bad donor interacts with a good recipient, and such donors are nevertheless assessed as good when they defect. For their definition, see Ref. (1).

Fig. S4 and **Fig. S5** are the analogues of **Fig. 3** and **Fig. 6** for the secondary-sixteen norms, respectively. These figures show that the secondary-sixteen norms perform much worse than the leading-eight norms in terms of both self-cooperation and

stability against ALLD. Moreover, the RIS rule does not effectively improve performance for this class of donor-assessment norms. This poor performance under private assessment is consistent with a previous study (2). One likely reason is that the secondary-sixteen are more sensitive to the interaction context, which makes it harder for players to synchronize their opinions under private assessment.

2. Stability conditions under public assessment

Murase and Hilbe (1) derive necessary and sufficient conditions for dual assessment norms to form a cooperative ESS (CESS) under public assessment. A norm is a CESS if it is stable against invasion by rare mutants, and if the average cooperation level approaches one as the error rate vanishes. In that case, one can show that the CESS conditions are as follows:

$$\begin{cases} R_1(G,G,C) = 1 \\ R_2(G,G,C) = 1 \\ R_1(G,B,D) + R_2(G,B,D) + R_1(B,G,C) + R_2(B,G,C) > 2 \\ S(G,G) = C \\ S(G,B) = D \\ S(B,G) = C \\ S(B,B) = \begin{cases} C & \text{when } b[R_1(B,B,C) - R_1(B,B,D)] > c[R_1(B,G,C) + R_2(G,B,D)] \\ D & \text{otherwise} \end{cases} \\ R_1(G,G,D) < 1 \\ R_1(B,G,C) > R_1(B,G,D) \\ \frac{b}{c} > \max\left\{ \frac{R_1(B,G,C)+R_2(G,B,D)}{R_1(G,G,C)-R_1(G,G,D)}, \frac{R_1(B,G,C)+R_2(G,B,D)}{R_1(B,G,C)-R_1(B,G,D)} \right\} \\ R_1(G,B,C) \le R_1(G,B,D) \quad \text{or} \quad \frac{b}{c} < \frac{R_1(B,G,C)+R_2(B,G,C)}{R_1(G,G,C)-R_1(G,G,D)} \end{cases} \quad [3]$$

These conditions hold for norms with $S(B,G) = C$. There are analogous conditions for norms with $S(B,G) = D$, but we omit them here for brevity. This is sufficient for our purposes, because below we apply the above conditions to the leading-eight norms, all of which prescribe $S(B,G) = C$.

In the following, we apply these conditions to the special case in which donors act and are assessed according to the leading-eight. The leading-eight share the following properties (see also **Table 1** in the main text),

$$R_1(G,G,C) = 1, \quad [4]$$
$$R_1(G,G,D) = 0, \quad [5]$$
$$R_1(G,B,D) = 1, \quad [6]$$
$$R_1(B,G,C) = 1, \quad [7]$$
$$R_1(B,G,D) = 0, \quad [8]$$
$$S(G,G) = C, \quad [9]$$
$$S(G,B) = D, \quad [10]$$
$$S(B,G) = C. \quad [11]$$

By substituting these values into Eq. [3], we obtain simplified necessary and sufficient conditions on R_2 for CESS:

$$\begin{cases} R_2(G,G,C) = 1 \\ R_2(G,B,D) + R_2(B,G,C) > 0 \\ b > [1 + R_2(G,B,D)]\, c. \end{cases} \quad [12]$$

This recovers Eq. [3] in the main text. In addition, the action rule must prescribe an appropriate action $S(B,B)$ when two bad individuals interact. This action depends on the b/c ratio, as specified in Eq. [3].

3. A recovery analysis from a single error for L6-RIS

For a detailed description of the recovery analysis, we consider the setup described in the main text. There is a population of size N, and initially everyone regards everyone as good, with a single exception: player 1 regards player 2 as bad. For the subsequent dynamics, we assume there are no further errors (i.e., the respective error rates μ are set to zero). We ask: How long does it take on average until everyone regards everyone as good again, assuming that all players adopt the L6-RIS norm?

In general, this question would require us to keep track of the entire $N \times N$ image matrix (which allows for 2^{N^2} possible states). However, as mentioned in the main text, one can reduce the state space by exploiting four observations:

1. The number of players about whom there is disagreement cannot increase. This is due to the fact that by the end of an interaction, all players agree on the reputation of the recipient (because they agree on the action that has been played). Therefore, if prior to the interaction, players disagreed on the reputation of the recipient, after the interaction there might be disagreements about the donor but not about the recipient.

2. All remaining players agree on the reputation they assign to any third party. This is because they all agree initially and assess all subsequent interactions in the same way (because they all apply the same norm and there are no further errors).

3. When chosen as a donor, a player will always regard herself as good after the interaction. (A donor is regarded as good insofar as she follows the prescribed action rule.)

4. When a disagreement about player 1 exists, player 1 will regard herself as good while the other players will regard player 1 as bad. Similarly, when a disagreement about a player j ($\neq 1$) exists, player 1 will regard j as bad while all other players will regard j as good. This is because the disagreement about a player j arises only when j is chosen as a donor, and the donor always regards herself as good after the interaction.

Based on these observations, we argued in the main text that we can characterize the current state of the population by a triplet (α, β, γ). We briefly summarize this description here. The first two entries represent whether player 1 is regarded as good and how many of the remaining players are good, both from the viewpoint of the remaining players. Hence, $\alpha \in \{0, 1\}$ and $\beta \in \{0, 1, \ldots, N-1\}$. The last entry $\gamma \in \{0, 1, -\}$ encodes where the disagreement exists (if there is any). If $\gamma = 0$, there is no disagreement between player 1 and the remaining players. If $\gamma = 1$, they disagree about player 1. Player 1 considers herself to be good whereas the others do not. (The opposite case never occurs as mentioned above. Thus, α must be 0 when $\gamma = 1$.) If $\gamma = -$, they disagree about one of the remaining players, say player j. While player 1 considers j as bad, the remaining players consider j as good. (The opposite case, where player 1 considers j as good while the remaining players consider j as bad, cannot arise as mentioned above.) Based on this notation, we ask: Given that initially we are in state (1,N-1,-), how long does it take to reach the full recovery state (1,N-1,0)?

To compute the system's recovery time analytically, we next compute the transition probabilities between different states. When the current state is $(\alpha, \beta, 0)$, i.e., there is no disagreement in opinions, no further disagreement can arise. Thus, the number of good players increases monotonically until reaching the absorbing state $(1, N-1, 0)$. In this scenario, a donor is always considered good after the interaction, regardless of the recipient's reputation. When a bad player is chosen as the donor, $\alpha+\beta$ increases by one, while it remains unchanged when a good player is chosen as the donor. Overall, $\alpha+\beta$ increases with probability $[N - (\alpha + \beta)]/N$. Otherwise, the state remains unchanged.

When the current state has a disagreement ($\gamma \neq 0$), the disagreement can either be resolved or persist. The transitions from those states are summarized in **Table S1**. Although there are minor differences in the transition probabilities for each state, the overall picture is as follows.

- When the player about whom there is a disagreement is chosen as the donor, the disagreement is resolved. This occurs with probability $1/N$ since there is only one such player.

- If the donor is perceived as bad from the perspective of the remaining players, the number of good players from the viewpoint of the remaining players, $\alpha+\beta$, increases by one. This occurs with probability $[N - (\alpha + \beta)]/N$.

- However, there is an exception to the previous point. When player 1 is chosen as the donor, player j is chosen as the recipient, and there is a disagreement about player j, $\alpha+\beta$ decreases. See the rows where the second column is (**1**, **G'**) in **Table S1**.

The first two events occur with probabilities of order $\Theta(1/N)$, while the last event occurs with probability $\Theta(1/N^2)$. Thus, the recovery time is approximately $\Theta(N)$, which is faster than that of the L6-base norm.

4. A recovery analysis from a random configuration for L6-RIS

We present another recovery analysis for L6-RIS, in which we start from a random configuration of opinions (instead of the specific configuration considered in the main text) to demonstrate that the recovery time remains controlled even in this case. Starting from this random configuration, we analyze the stochastic dynamics until the system reaches the full recovery state. As shown below, the system drifts toward the full recovery state in expectation, and the expected recovery time scales as $\Theta(N \log N)$, whereas the recovery time for L6-base diverges.

The state of a player is characterized by the opinions of other players about that player. When a focal player is unanimously regarded as good by all other players, we refer to the focal player as being in **G** state. Similarly, when a focal player is unanimously regarded as bad, we refer to the focal player as being in **B** state. When there is a disagreement about the focal player, we refer to the focal player as being in **M** state. In this analysis, half of the players in the population consider a focal player in the **M** state as good, while the other half consider the focal player as bad. (We need not consider other fractions of opinions, because disagreement about a player arises only when a **M** player is chosen as the recipient; in that case, opinions about the donor are split into two groups of equal size.) The state of the image matrix can be characterized by the number of players in each state, which we denote as $n_{\mathbf{G}}$, $n_{\mathbf{B}}$, and $n_{\mathbf{M}}$ for the number of players in **G**, **B**, and **M** states, respectively, with $n_{\mathbf{G}} + n_{\mathbf{B}} + n_{\mathbf{M}} = N$. Starting from a random configuration $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}}) = (0, 0, N)$, we analyze the stochastic dynamics until reaching the full recovery state $(N, 0, 0)$.

Table S2 summarizes the transition probabilities of the Markov process from $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$. As shown in the table, the number of disagreements, $n_{\mathbf{M}}$, is non-increasing. This is because the players agree on the recipient's reputation after the interaction. Similarly, the number of players in **G** state, $n_{\mathbf{G}}$, increases in expectation as long as $n_{\mathbf{G}} < N$. Thus, the system

approaches the full recovery state $(N,0,0)$ in expectation. This is in stark contrast to the L6-base norm, where the number of disagreements never decreases, and the system can get stuck in a state with many disagreements indefinitely.

The transition table also implies that the full recovery state $(N,0,0)$ is the only absorbing state, provided $N \geq 2$. Clearly, once the system is in state $(N,0,0)$, every possible interaction leaves the state unchanged. Conversely, consider any other state $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$. If $n_{\mathbf{M}} = 0$ and $n_{\mathbf{B}} > 0$, choosing a $\mathbf{B}$ player as the donor has positive probability. In that case, the donor changes from $\mathbf{B}$ to $\mathbf{G}$, so the system moves to $(n_{\mathbf{G}}+1, n_{\mathbf{B}}-1, 0)$. If $n_{\mathbf{M}} > 0$, there are two cases. When $n_{\mathbf{M}} \geq 2$, choosing two $\mathbf{M}$ players has positive probability, and the recipient's disagreement is resolved, leading to either $(n_{\mathbf{G}}+1, n_{\mathbf{B}}, n_{\mathbf{M}}-1)$ or $(n_{\mathbf{G}}, n_{\mathbf{B}}+1, n_{\mathbf{M}}-1)$. When $n_{\mathbf{M}} = 1$, at least one of $n_{\mathbf{G}}$ or $n_{\mathbf{B}}$ is positive. If $n_{\mathbf{G}} > 0$, the transition $(\mathbf{G},\mathbf{M}) \to (\mathbf{M},\mathbf{B})$ changes the state to $(n_{\mathbf{G}}-1, n_{\mathbf{B}}+1, n_{\mathbf{M}})$ with positive probability. If $n_{\mathbf{B}} > 0$, the transition $(\mathbf{B},\mathbf{M}) \to (\mathbf{M},\mathbf{G})$ changes the state to $(n_{\mathbf{G}}+1, n_{\mathbf{B}}-1, n_{\mathbf{M}})$ with positive probability. Thus, every state other than $(N,0,0)$ has a positive probability of leaving, which proves that $(N,0,0)$ is the unique absorbing state.

This transition structure also allows us to derive the asymptotic scaling of the recovery time from the random initial condition. Let T_N be the expected time to reach $(N,0,0)$ from $(0,0,N)$. First, consider the time it takes until all disagreements have disappeared, i.e., until $n_{\mathbf{M}} = 0$. From any state with $n_{\mathbf{M}} = m$, the number of $\mathbf{M}$ players decreases by one exactly when the donor is a $\mathbf{M}$ player. According to **Table S2**, the corresponding total probability is

$$\frac{n_{\mathbf{G}} n_{\mathbf{M}} + n_{\mathbf{B}} n_{\mathbf{M}} + n_{\mathbf{M}}(n_{\mathbf{M}}-1)}{N(N-1)} = \frac{m}{N}. \quad [13]$$

Therefore, the expected time to reduce $n_{\mathbf{M}}$ from m to $m-1$ is N/m. Starting from $n_{\mathbf{M}} = N$, the expected time until all disagreements are resolved is

$$\sum_{m=1}^{N} \frac{N}{m} = N H_N, \quad [14]$$

where $H_N = \sum_{m=1}^{N} 1/m$ is the N-th harmonic number. Since full recovery cannot occur before all disagreements have disappeared, this gives the lower bound

$$T_N \geq N H_N. \quad [15]$$

Once $n_{\mathbf{M}} = 0$, the only remaining non-recovered players are in state $\mathbf{B}$. If there are $n_{\mathbf{B}} = b$ such players, the probability that a $\mathbf{B}$ player is chosen as the donor is b/N, and this event reduces b by one. Hence, the expected time to convert all remaining $\mathbf{B}$ players into $\mathbf{G}$ players is at most

$$\sum_{b=1}^{N} \frac{N}{b} = N H_N. \quad [16]$$

Combining the two bounds, we obtain

$$N H_N \leq T_N \leq 2 N H_N. \quad [17]$$

This scaling is consistent with our simulation results shown in **Fig. S6**.

5. Local stability against behavioral mutants

In the main text, we have focused on the stability of the resident norm against ALLC and ALLD mutants. Although ALLC and ALLD are two extreme action-rule mutants, one may also ask whether the resident norm is stable against other behavioral mutants. To strengthen our results, we therefore tested the evolutionary stability of the resident norm against all possible mutants that differ in their action rules, while keeping the assessment rules fixed to those of the resident norm.

Fig. S7 shows the payoffs of each resident norm and the corresponding mutants that differ from it in their action rule. We introduced a single mutant player into a resident population and computed the long-term average payoffs for $b/c = 2$. As shown in the figure, L6-base and L8-base are vulnerable to several behavioral mutants, including ALLD. By contrast, L6-RIS and L8-RIS are robust against all action-rule mutants considered here. Thus, the stability of L6-RIS and L8-RIS is not limited to ALLC or ALLD mutants but extends to the full set of mutants with alternative action rules under the same assessment rules.

We note, however, that stability against all behavioral mutants does not imply stability against every possible mutant. For example, mutants with different assessment rules may obtain higher payoffs than the resident norm and thus invade. Such an example was found in a previous study (2), where L2 was shown to be vulnerable to invasion by a mutant with L1 assessment rules. In general, it is difficult to prove that a resident norm is stable against all possible mutants under private assessment, because the optimal action depends on the current image matrix.

6. Imperfect observations

The results reported in the main text and in the previous sections are based on the assumption that all players observe the same interaction. Here, we consider imperfect observations: each player observes a given interaction with probability q, independently of other players. **Fig. S8** shows how the self-cooperation level and the critical b/c ratio above which the resident norm is stable against ALLD mutants depend on q for L6-base and L6-RIS.

As shown in the figure, L6-RIS continues to support cooperation when observations are imperfect, provided the observation probability is sufficiently large. Both the self-cooperation level and the critical b/c ratio change smoothly as q decreases. By

contrast, the self-cooperation level of L6-base is independent of q, and L6-base does not sustain cooperation for any value of q. As q approaches zero, the self-cooperation level under L6-RIS approaches 0.5, and the critical b/c ratio diverges. This limiting behavior is consistent with a previous study (3), which showed that no social norm can sustain cooperation when players' opinions are uncorrelated, as occurs when $q = 1/N$.

7. Comprehensive comparison of all assessment rules

The main text focuses on the leading-eight R_1 rules and compares various R_2 rules for assessing recipients. Here, we provide a comprehensive comparison across all possible combinations of deterministic R_1 and R_2 rules, using large-scale computational resources. Because there are $2^8 = 256$ possible deterministic rules for each of R_1 and R_2, this yields $256 \times 256 = 65,536$ possible (R_1, R_2) combinations. For all combinations, we fixed the action rule S to be the DISC rule, which prescribes cooperation if and only if the recipient is deemed good.

We evaluated each combination using three metrics: the self-cooperation level, the equilibrium fraction in evolutionary competition among the focal norm, ALLC, and ALLD, and local stability against mutants with alternative action rules. First, we calculated the self-cooperation level by running simulations for a monomorphic population. Second, we computed the equilibrium fraction of the focal norm in evolutionary competition among the focal norm, ALLC, and ALLD, using the same method as in the main text. These first two metrics are the same as those used in **Fig. 6** in the main text. In addition, we computed the payoffs of all possible mutants with alternative action rules, as in **Fig. S7**. We introduced a single mutant into a resident population of $N - 1$ individuals and computed the long-term average payoffs for $b/c = 2$. We judged the resident norm to be locally stable against behavioral mutants if no mutant had a payoff that exceeded the resident payoff by more than 0.02. We used this tolerance to account for numerical errors in the payoff calculations; since some mutants differ only slightly from the resident action rule, their payoffs can be very close to that of the resident norm, and thus we cannot judge stability based on a strict inequality.

Fig. S9 summarizes the results of this comprehensive comparison. The horizontal and vertical axes show the self-cooperation level and the equilibrium fraction in the triadic evolutionary competition, respectively. The colors of the symbols indicate whether the norm is locally stable against behavioral mutants. Blue symbols in the upper-right corner correspond to norms that perform well according to all three metrics. As shown in the figure, leading-eight norms with $R_2 = \text{RIS}$, which are highlighted as green squares, are among the best-performing norms. In particular, L2v-RIS, L5-RIS, L6-RIS, and L8-RIS perform well in terms of all metrics (L1v and L2v are considered here instead of L1 and L2 because the action rule is fixed to DISC).

We further extended this analysis by sweeping over all possible deterministic action rules S for each combination of R_1 and R_2. The number of possible deterministic action rules is $2^4 = 16$, so this yields $65,536 \times 16 = 1,048,576$ possible (R_1, R_2, S) combinations. We excluded norms that are equivalent to each other under $G \leftrightarrow B$ symmetry. This approximately halves the number of norms, but the conclusions remain unchanged. We filtered the norms using the same three metrics as before: (1) self-cooperation level greater than 0.8, (2) equilibrium fraction in the triadic evolutionary competition greater than 0.95, and (3) local stability against behavioral mutants with a tolerance of 0.02. We found 61 norms that satisfy all three criteria; see **Table S3**. They exhibit several interesting patterns. First, all of them have leading-eight R_1 rules (either L2, L5, L6, or L8) and DISC-like action rules; most use the DISC rule, whereas several use its variant with $S(B, B) = C$. Second, all of them require some form of dual reputation updating. Their R_2 rules are either RIS, combinations of the RIS and base rules (i.e., if there is any dual reputation update, it is based on RIS), or variants that differ from those combinations in only one or two entries. Third, RIS is the only R_2 rule that satisfies all three criteria with all four leading-eight R_1 rules, L2, L5, L6, and L8. These results provide some ex-post justification for our focus on Lx-RIS norms in the main text, while also showing that dual reputation updates can give rise to a broader class of cooperative and evolutionarily stable norms.

References

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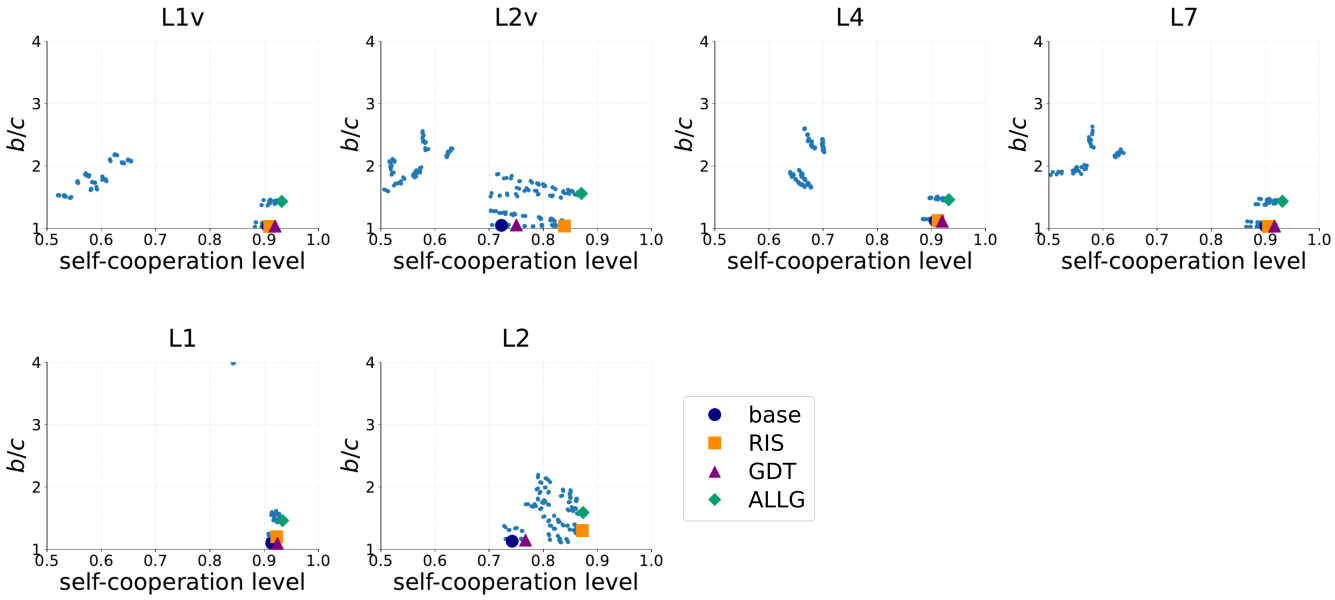


Fig. S1. Cooperation levels and stability against ALLD across recipient-assessment rules for the remaining leading-eight norms. We pair each donor-assessment rule, $R_1 \in \{\text{L1, L1v, L2, L2v, L4, L7}\}$, with all possible deterministic recipient-assessment rules, R_2. For each pair, we consider a homogeneous population in which everyone applies the respective norm. For that population, we compute its self-cooperation level (x-axis) and the critical b/c ratio above which the norm is stable against ALLD (y-axis). Points toward the bottom-right corner of each panel correspond to norms that perform well with respect to both dimensions. The symbols for norms with $R_2 =$ base, RIS, GDT, and ALLG, and the parameters are the same as in **Fig. 3**.

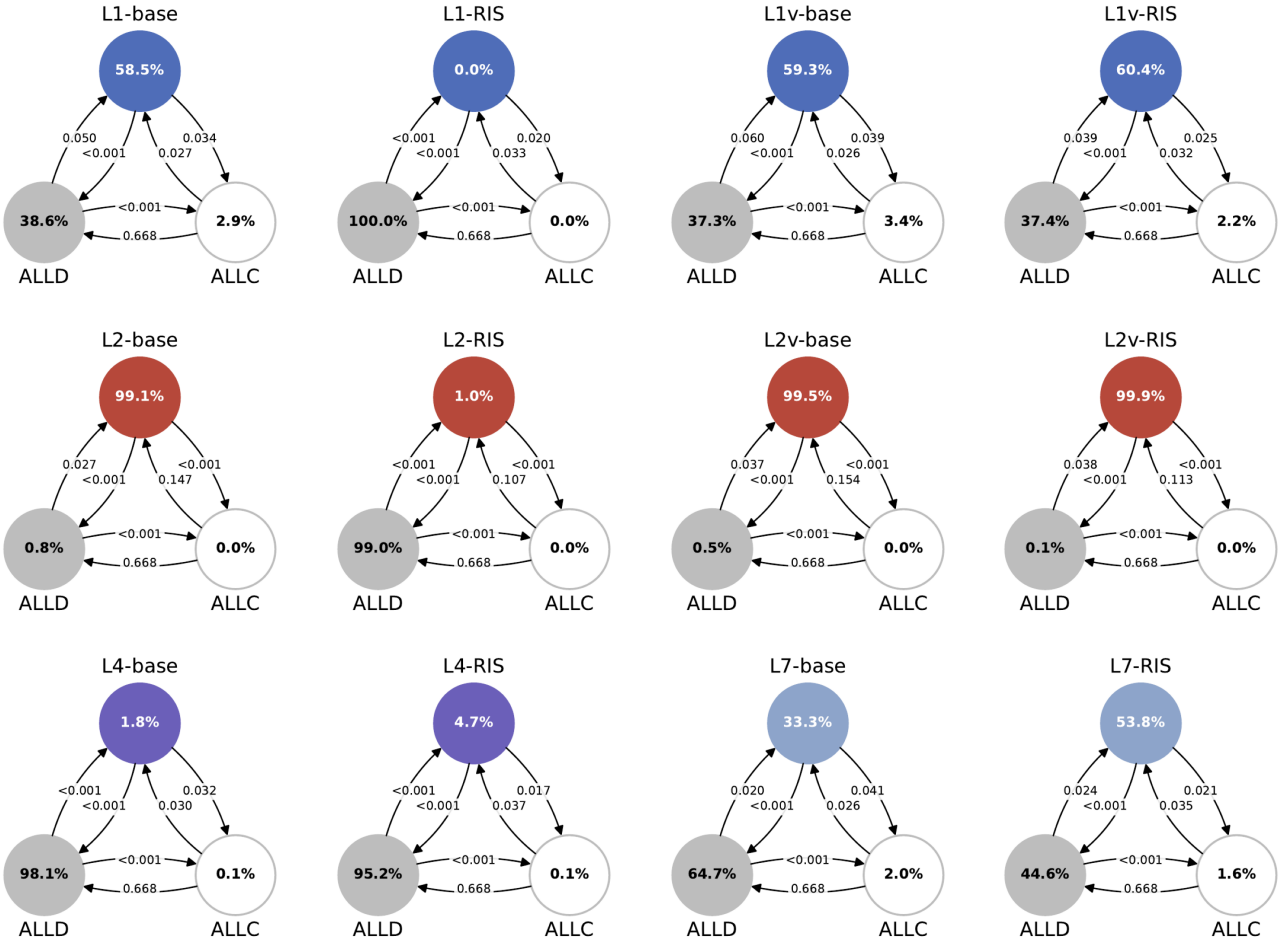


Fig. S2. Evolutionary competition among the focal norm, ALLC, and ALLD for the remaining leading-eight norms. We compare the base and RIS rules for recipient assessment across six donor-assessment rules: L1, L1v, L2, L2v, L4, and L7. The numbers near arrows indicate transition probabilities, and the equilibrium fractions are shown in the circles. Parameters are the same as in **Fig. 5**.

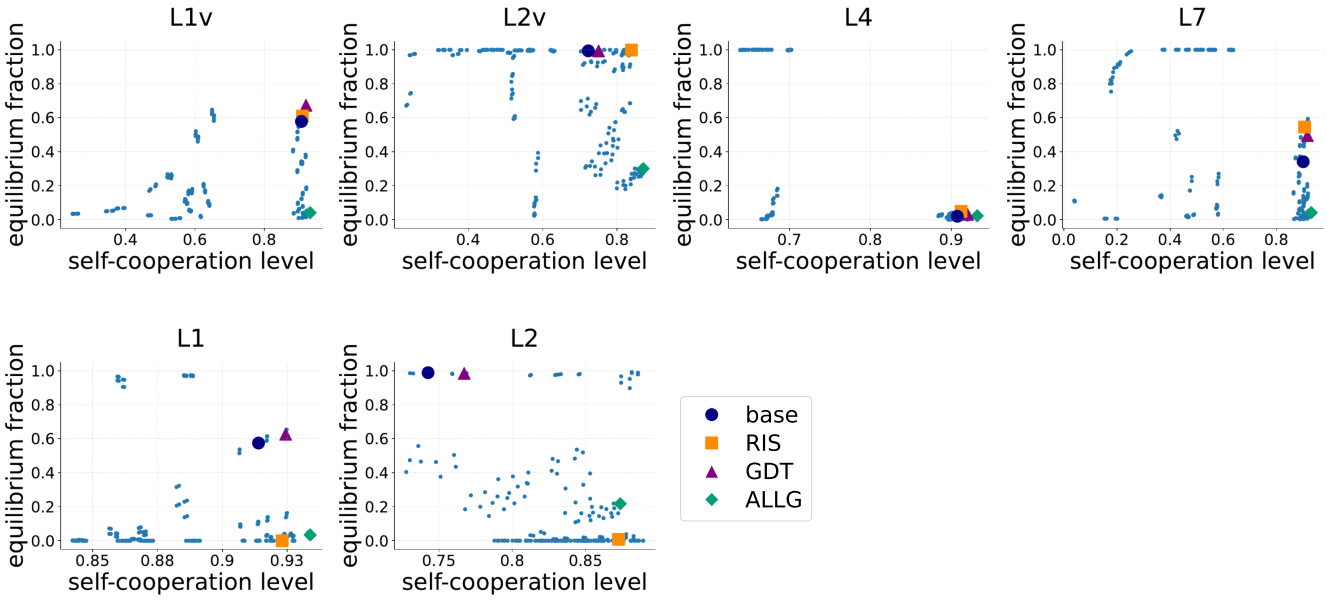


Fig. S3. Cooperation levels and evolutionary abundance across recipient-assessment rules for the remaining leading-eight norms. We sweep over all possible deterministic R_2 rules while fixing R_1 and S to be the same as L1, L1v, L2, L2v, L4, and L7. The horizontal axis shows self-cooperation levels in monomorphic populations, and the vertical axis shows the equilibrium fraction of the focal norm in evolutionary competition among the focal norm, ALLC, and ALLD. Points toward the top-right corner of each panel correspond to norms that perform well with respect to both dimensions. The symbols for norms with $R_2 =$ base, RIS, GDT, and ALLG, and the parameters are the same as in **Fig. 6**.

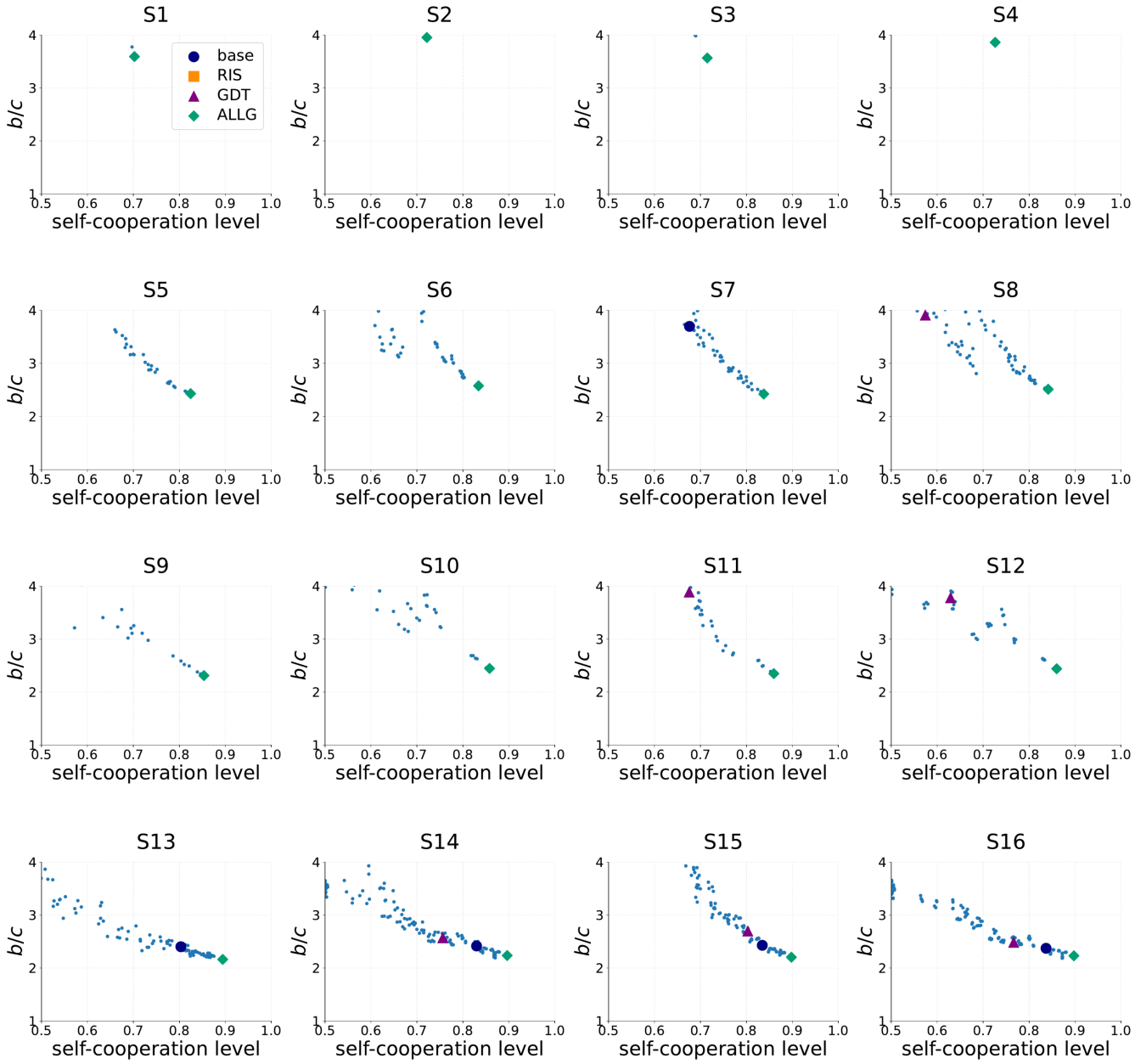


Fig. S4. Cooperation levels and stability against ALLD across recipient-assessment rules for the secondary-sixteen norms. We sweep over all possible deterministic recipient-assessment rules, R_2, while fixing S and R_1 to match the secondary-sixteen norms. For each resulting norm, we compute its self-cooperation level (x-axis) and the critical b/c ratio above which the resident norm is stable against ALLD mutants (y-axis). Points toward the bottom-right region correspond to norms that perform well with respect to both dimensions. The symbols for norms with $R_2 =$ base, RIS, GDT, and ALLG, and the parameters are the same as in **Fig. 3**.

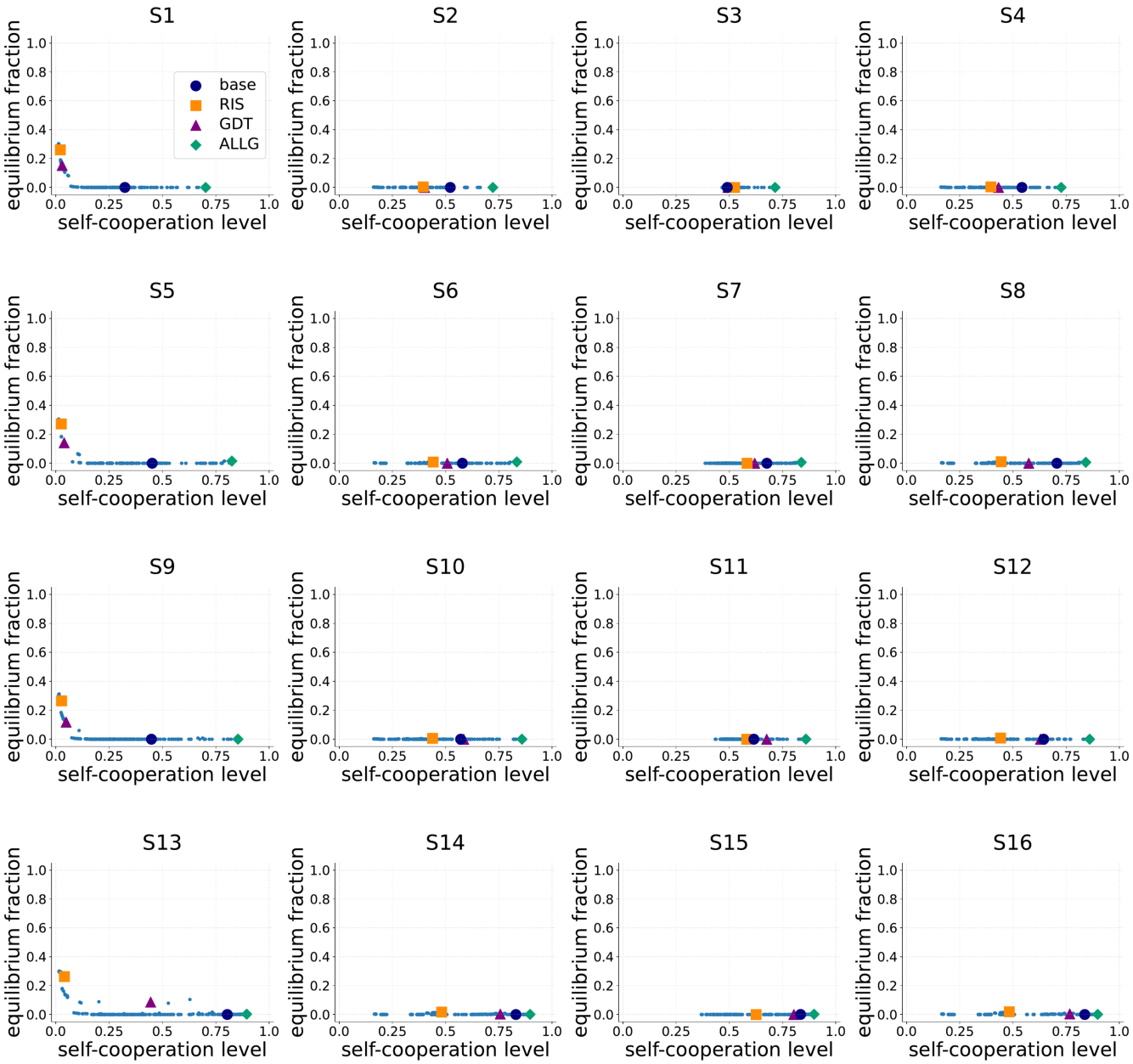


Fig. S5. Cooperation levels and evolutionary abundance across recipient-assessment rules for the secondary-sixteen norms. We sweep over all possible deterministic recipient-assessment rules, R_2, while fixing S and R_1 to match the secondary-sixteen norms. The horizontal axis shows self-cooperation levels in monomorphic populations, and the vertical axis shows the equilibrium fraction of the focal norm in evolutionary competition among the focal norm, ALLC, and ALLD. Points toward the top-right region correspond to norms that perform well with respect to both dimensions. The symbols for norms with $R_2 =$ base, RIS, GDT, and ALLG, and the parameters are the same as in **Fig. 6**.

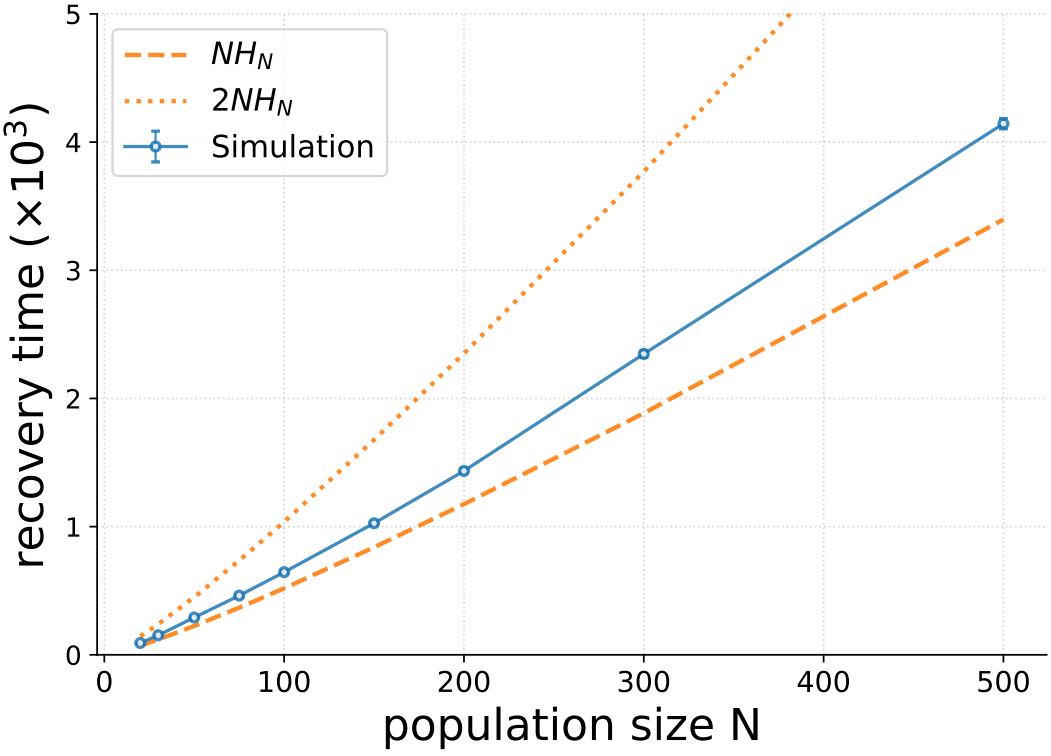


Fig. S6. Recovery time from a random configuration for L6-RIS. The plot shows the mean recovery time from the random initial condition $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}}) = (0, 0, N)$ to the full recovery state $(N, 0, 0)$ as a function of the population size N. For each N, we ran 300 Monte Carlo simulations of the stochastic process and report the mean recovery time. The simulation results are consistent with the analytical scaling $NH_N \leq T_N \leq 2NH_N$, where H_N is the N-th harmonic number.

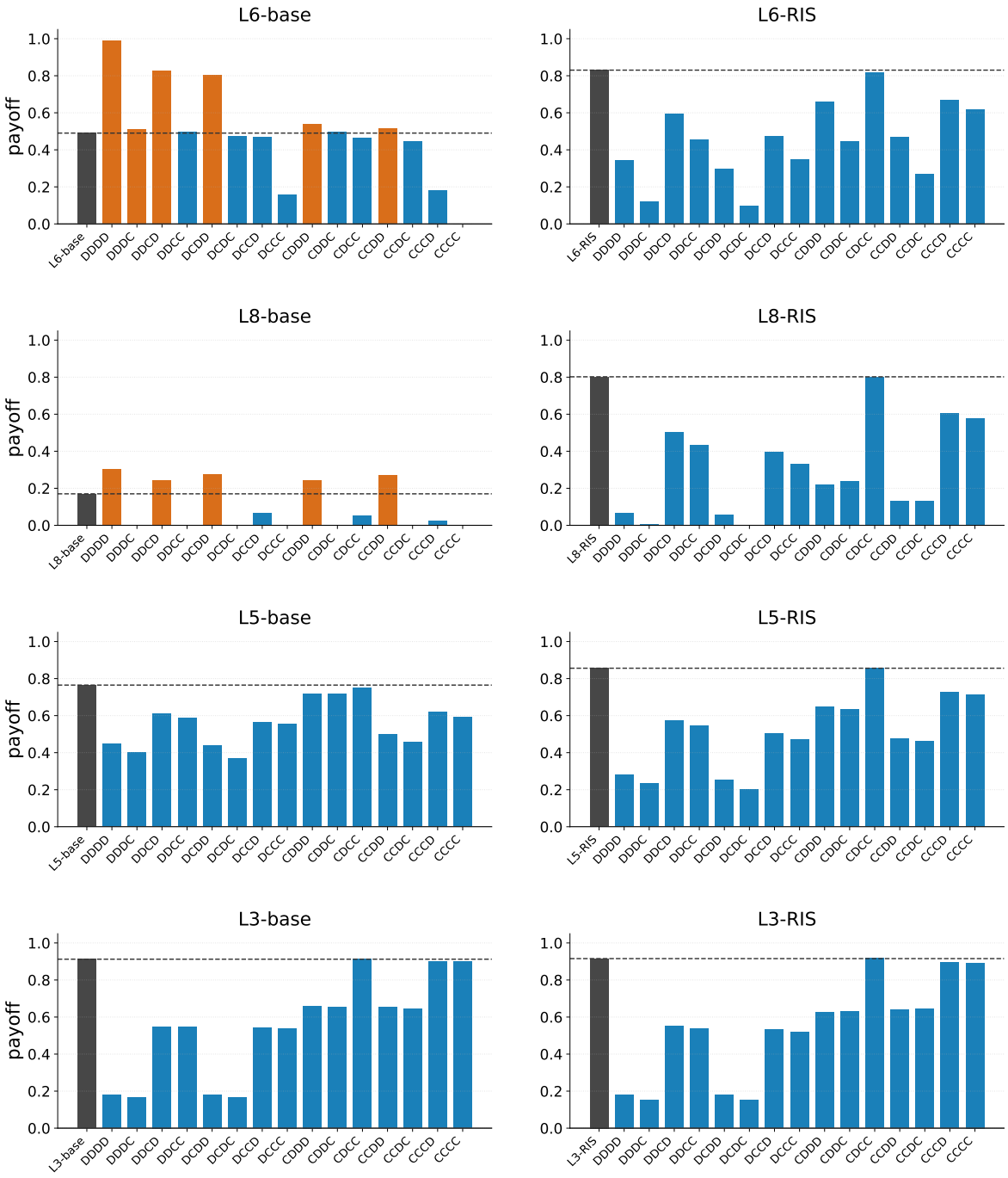


Fig. S7. Payoffs of behavioral mutants. Each panel shows the payoff of the resident norm and the payoffs of mutants that differ from the resident norm only in their action rule. We introduced a single mutant into a resident population and computed the long-term average payoffs for $b/c = 2$. The leftmost bar shows the payoff of the resident norm, and the remaining bars show the payoffs of the corresponding behavioral mutants. The horizontal-axis labels indicate the mutant action rule. Each action rule is represented by a four-letter string, where the letters specify the mutant's action when the (donor, recipient) pair is (G, G), (G, B), (B, G), and (B, B), respectively. The horizontal dashed line indicates the resident payoff, and mutants with payoffs above $\pi_{\text{resident}} + 0.02$ are highlighted in orange. Bars for mutants with negative payoffs are omitted for better visualization. The baseline parameters are the same as those used in the main text.

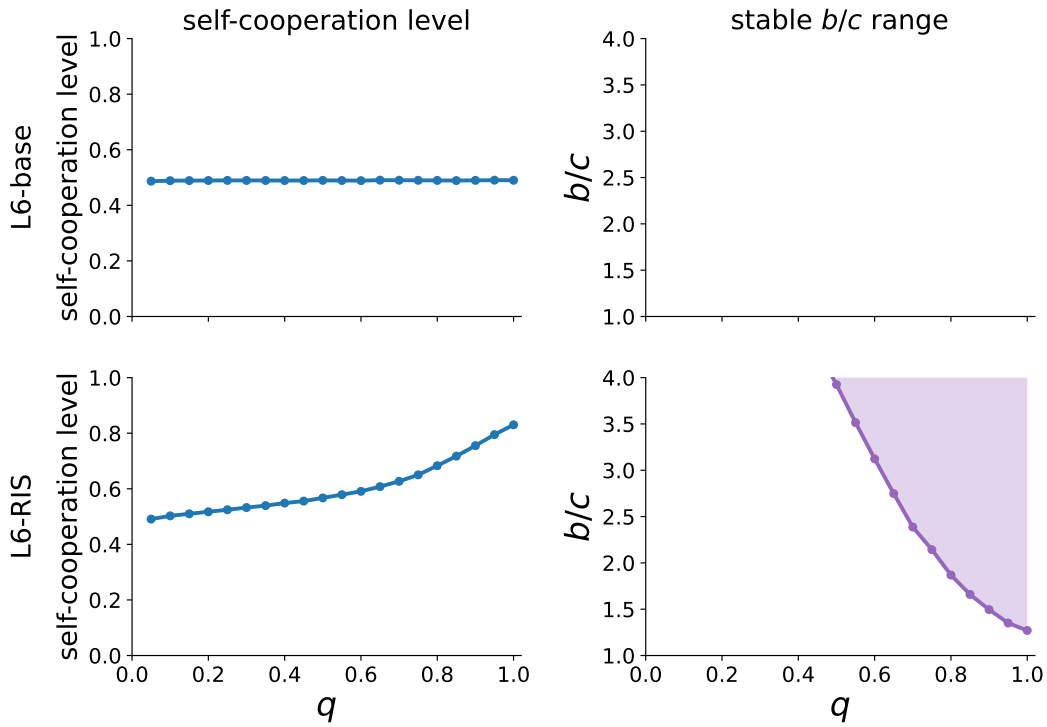


Fig. S8. Self-cooperation level and stability against ALLD under imperfect observations. The left panels show the self-cooperation level as a function of the observation probability q. The right panels show the critical b/c ratio above which the resident norm is stable against ALLD mutants, when such a stable range exists. The top panels show results for L6-base, and the bottom panels show results for L6-RIS. The baseline parameters are the same as those used in the main text, except that the observation probability q is varied.

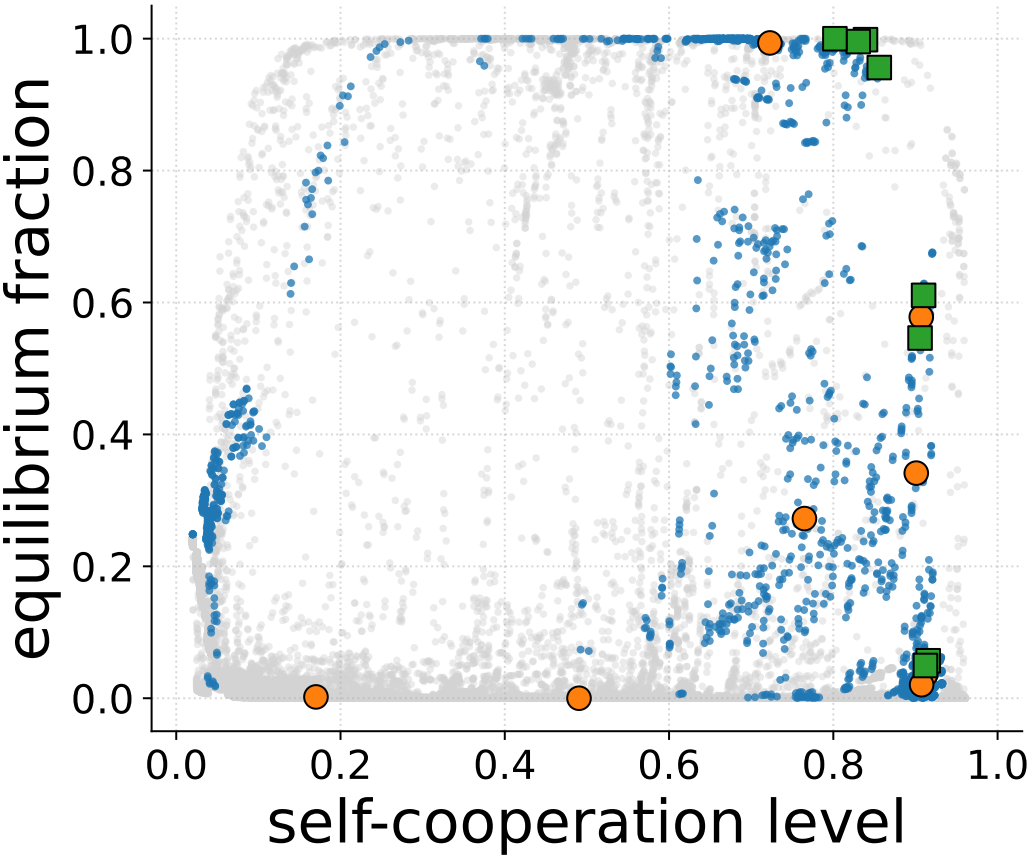


Fig. S9. Comprehensive comparison of all norms. For each of the 65,536 possible combinations of R_1 and R_2 rules, we computed the self-cooperation level in a monomorphic population, shown on the horizontal axis, and the equilibrium fraction of the focal norm in evolutionary competition among the focal norm, ALLC, and ALLD, shown on the vertical axis. The action rule S was fixed to be the DISC rule for all norms. In addition, we computed the payoffs of all possible mutants with alternative action rules and judged the resident norm to be locally stable against behavioral mutants if no mutant had a payoff that exceeded the resident payoff by more than 0.02. Locally stable norms are plotted as blue circles, whereas locally unstable norms are plotted as gray circles. The Lx-base norms are highlighted as orange circles, and the Lx-RIS norms are highlighted as green squares. Because the action rule is fixed to DISC, L1v and L2v are shown instead of L1 and L2. All Lx-RIS norms are locally stable against behavioral mutants, whereas some Lx-base norms are not. Among Lx-RIS norms, L2v-RIS, L5-RIS, L6-RIS, and L8-RIS perform well in terms of both cooperation level and evolutionary stability.

current state	(donor, recipient)	probability ($\times N(N-1)$)	next state
$(0,\beta,1)$	(**1**, **G**)	β	$(1,\beta,0)$
	(**1**, **B**)	$N-1-\beta$	$(1,\beta,0)$
	(**G**, **1**)	β	$(0,\beta,-)$
	(**G**, **G**)	$\beta(\beta-1)$	$(0,\beta,1)$
	(**G**, **B**)	$\beta(N-1-\beta)$	$(0,\beta,1)$
	(**B**, **1**)	$N-1-\beta$	$(0,\beta+1,-)$
	(**B**, **G**)	$(N-1-\beta)\beta$	$(0,\beta+1,1)$
	(**B**, **B**)	$(N-1-\beta)(N-2-\beta)$	$(0,\beta+1,1)$
$(1,\beta,-)$	(**1**, **G**)	$\beta-1$	$(1,\beta,-)$
	(**1**, **G'**)	1	$(0,\beta-1,1)$
	(**1**, **B**)	$N-1-\beta$	$(1,\beta,-)$
	(**G**, **1**)	$\beta-1$	$(1,\beta,-)$
	(**G**, **G**)	$(\beta-1)(\beta-2)$	$(1,\beta,-)$
	(**G**, **G'**)	$\beta-1$	$(1,\beta,-)$
	(**G**, **B**)	$(\beta-1)(N-1-\beta)$	$(1,\beta,-)$
	(**G'**, **1**)	1	$(1,\beta,0)$
	(**G'**, **G**)	$\beta-1$	$(1,\beta,0)$
	(**G'**, **B**)	$N-1-\beta$	$(1,\beta,0)$
	(**B**, **1**)	$N-1-\beta$	$(1,\beta+1,-)$
	(**B**, **G**)	$(N-1-\beta)(\beta-1)$	$(1,\beta+1,-)$
	(**B**, **G'**)	$N-1-\beta$	$(1,\beta+1,-)$
	(**B**, **B**)	$(N-1-\beta)(N-2-\beta)$	$(1,\beta+1,-)$
$(0,\beta,-)$	(**1**, **G**)	$\beta-1$	$(1,\beta,-)$
	(**1**, **G'**)	1	$(0,\beta-1,1)$
	(**1**, **B**)	$N-1-\beta$	$(1,\beta,-)$
	(**G**, **1**)	$\beta-1$	$(0,\beta,-)$
	(**G**, **G**)	$(\beta-1)(\beta-2)$	$(0,\beta,-)$
	(**G**, **G'**)	$\beta-1$	$(0,\beta,-)$
	(**G**, **B**)	$(\beta-1)(N-1-\beta)$	$(0,\beta,-)$
	(**G'**, **1**)	1	$(0,\beta,0)$
	(**G'**, **G**)	$\beta-1$	$(0,\beta,0)$
	(**G'**, **B**)	$N-1-\beta$	$(0,\beta,0)$
	(**B**, **1**)	$N-1-\beta$	$(0,\beta+1,-)$
	(**B**, **G**)	$(N-1-\beta)(\beta-1)$	$(0,\beta+1,-)$
	(**B**, **G'**)	$N-1-\beta$	$(0,\beta+1,-)$
	(**B**, **B**)	$(N-1-\beta)(N-2-\beta)$	$(0,\beta+1,-)$

Table S1. Transition probabilities for the L6-RIS norm when opinions disagree. In the second column, **1** indicates player 1, **G** indicates a remaining player who is considered good by the remaining players, and **B** indicates a remaining player considered bad by the remaining players. **G'** indicates a remaining player who is considered good by the remaining players but bad by player 1. The third column shows the transition probabilities multiplied by $N(N-1)$ for better readability.

current (donor, recipient) state	next (donor, recipient) state	probability ($\times N(N-1)$)	next system state
(**G**, **G**)	(**G**, **G**)	$n_{\mathbf{G}}(n_{\mathbf{G}}-1)$	$(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$
(**B**, **G**)	(**G**, **G**)	$n_{\mathbf{G}}n_{\mathbf{B}}$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}-1, n_{\mathbf{M}})$
(**M**, **G**)	(**G**, **G**)	$n_{\mathbf{G}}n_{\mathbf{M}}$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}, n_{\mathbf{M}}-1)$
(**G**, **B**)	(**G**, **B**)	$n_{\mathbf{G}}n_{\mathbf{B}}$	$(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$
(**B**, **B**)	(**G**, **B**)	$n_{\mathbf{B}}(n_{\mathbf{B}}-1)$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}-1, n_{\mathbf{M}})$
(**M**, **B**)	(**G**, **B**)	$n_{\mathbf{B}}n_{\mathbf{M}}$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}, n_{\mathbf{M}}-1)$
(**G**, **M**)	(**M**, **G**)	$n_{\mathbf{G}}n_{\mathbf{M}}/2$	$(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$
(**G**, **M**)	(**M**, **B**)	$n_{\mathbf{G}}n_{\mathbf{M}}/2$	$(n_{\mathbf{G}}-1, n_{\mathbf{B}}+1, n_{\mathbf{M}})$
(**B**, **M**)	(**M**, **G**)	$n_{\mathbf{B}}n_{\mathbf{M}}/2$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}-1, n_{\mathbf{M}})$
(**B**, **M**)	(**M**, **B**)	$n_{\mathbf{B}}n_{\mathbf{M}}/2$	$(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$
(**M**, **M**)	(**M**, **G**)	$n_{\mathbf{M}}(n_{\mathbf{M}}-1)/2$	$(n_{\mathbf{G}}+1, n_{\mathbf{B}}, n_{\mathbf{M}}-1)$
(**M**, **M**)	(**M**, **B**)	$n_{\mathbf{M}}(n_{\mathbf{M}}-1)/2$	$(n_{\mathbf{G}}, n_{\mathbf{B}}+1, n_{\mathbf{M}}-1)$

Table S2. Transition probabilities for the L6-RIS norm from a random configuration. Transition probabilities for the Markov process from $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$. The first two columns indicate the current and next states of the donor and recipient, respectively. The third column shows the transition probabilities multiplied by $N(N-1)$ for better readability. The last column indicates the next state of the system, i.e., the next values of $(n_{\mathbf{G}}, n_{\mathbf{B}}, n_{\mathbf{M}})$.

R_1	Action rule S	R_2 prescription	Count
L2	DISC	G**BG*GB	8
L2	DISC	G**GGBGB	4
L2	DISC	G*GBB*GB	4
L2	DISC	G*GGBBGB	2
L2	DISC with $S(B,B)=C$	G**B*GGB	8
L5	DISC	G*BB*BGB	4
L5	DISC	GBBB*BGG	2
L5	DISC	G*GB*BBB	4
L5	DISC	GBGBBB*G	2
L5	DISC	GBGB*BGB	2
L5	DISC	G*GBGBBG	2
L6	DISC	G**B*BG*	16
L8	DISC	G*GBGBGB	2
L8	DISC with $S(B,B)=C$	GGGBGGGB	1

Table S3. Comprehensive comparison of all norms that satisfy all three criteria. List of norms (R_1, R_2, S) that satisfy all three criteria: (1) self-cooperation level > 0.8, (2) equilibrium fraction in the triadic evolutionary competition > 0.95, and (3) local stability against behavioral mutants with a tolerance of 0.02. The first column shows the donor-assessment rule R_1, the second column shows the action rule S, and the third column shows the recipient-assessment rule R_2. The donor-assessment rule is one of L2, L5, L6, or L8, and the action rule is either the discriminator rule (DISC) or a variant of DISC with $S(B,B)=C$. The R_2 prescription is compressed into an eight-character string ordered as $R_2(G,G,C)$, $R_2(G,G,D)$, $R_2(G,B,C)$, $R_2(G,B,D)$, $R_2(B,G,C)$, $R_2(B,G,D)$, $R_2(B,B,C)$, and $R_2(B,B,D)$. Each character gives the assigned reputation for that entry; $*$ is an independent wildcard within the row. Underlined characters indicate entries that match the RIS rule. The rightmost column shows the number of norms. The parameters used in the evaluation are the same as those in the main text.